

Evaluating ESM3 Representations and Whole-Sequence Generative Models for HIV Broadly Neutralizing Antibody Modeling

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Abstract

We evaluate whether representations from the ESM3 protein foundation model transfer to sequence-only modeling of HIV broadly neutralizing antibodies (bnAbs), and if domain adaptation via parameter-efficient fine-tuning alters this transfer. We partition heavy-chain sequences into metadata-derived groups and evaluate frozen ESM3 and domain-adapted representations on classification and regression tasks. Frozen ESM3 representations carry bnAb neutralization-breadth signal and produce higher classification point estimates than baseline features, although paired group-bootstrap intervals include zero. The evaluated OAS LoRA adaptation run does not improve downstream performance. We also evaluate a whole-sequence categorical variational autoencoder (VAE). Its out-of-fold residue reconstruction accuracy is 64.2%. In a four-triple VRC34 maturation test, latent midpoints recover the known intermediate in two triples, but decoded midpoints remain farther from the target sequence than the nearer raw endpoint. These findings support frozen ESM3 as a viable exploratory representation for antibody modeling, while highlighting the importance of robust validation.

1. Introduction

HIV-1 broadly neutralizing antibodies (bnAbs) are specialized immune system proteins capable of neutralizing a wide range of viral variants by binding to comparatively conserved regions on the viral envelope glycoprotein [18, 26]. Many HIV-1 bnAbs arise after prolonged affinity maturation and carry unusually high levels of somatic hypermutation, B-cell receptor genes undergo mutation and selection [18, 26]. This process results in complex clonal lineages (Figure 1 summarizes mutations in selected members of one family)—groups of antibodies that share a common genetic ancestor (the Unmutated Common Ancestor, or UCA) but diverge in sequence and function as they ma-

A							
Name	Putative V gene	Putative J gene	CDR3 length (aa)	CDR3 Sequence (aa)	V _H J _H % Mut (nt)	V _H J _H % Mut (aa)	Insertions/Deletions (aa)
3H	IGHV4-59	IGHJ6*03	26	CARAQQGKRITGVMSFGFFYYNDVW	5%	11%	
4H				CARAQQGKRITGVMSFGELFYNDVW	6%	12%	
6H				CARAQRAQRIYGTISLGNFFDLYNDVW	19%	28%	
9H				CARTQHGRITGVAFNEMFTFYNDVW	12%	18%	
10H				CATTKHSRRIYGVAFNEMFTFYNDVW	16%	25%	
13H				CATTKHSRRIYGVAFNEMFTFYNDVW	22%	27%	
22H				CATTKHSRRIYGVAFNEMFTFYNDVW	23%	31%	
32H				CATTARRGRIYGVMSFGFFYYNDVW	11%	17%	
36H				CATTARRGRIYGVMSFGFFYYNSHDVW	15%	22%	
PGT121				CARTLHGRRIYGVAFNEMFTFYNDVW	18%	23%	
PGT122	CATTKHSRRIYGVAFNEMFTFYNDVW	19%	25%				
PGT123	CARALHGRRIYGVAFNEMFTFYNDVW	20%	27%				
PGT124	CATTARRGRIYGVMSFGFFYYNDVW	15%	21%				
PGT133	CARAQQGKRITGVMSFGFFYYNDVW	17%	25%				
PGT134	CARAQQGKRITGVMSFGELFYNDVW	17%	24%				

B

Name	Putative V gene	Putative J gene	CDR3 length (aa)	CDR3 Sequence (aa)	V _H J _H % Total Mut (nt)	V _H J _H % Total Mut (aa)	Insertions/Deletions (aa)
3L	IGHV3-21	IGHJ3*02	15	DYTCMWDSSRLNWF	9%	17%	+3 (FR3)
4L				DYTCIWDSSRPTKWF	16%	29%	-7 (FR1)/ +3 (FR3)
7L				DYTCIWDSSRPTKWF	21%	35%	-7 (FR1)/ +3 (FR3)
8L				ATYCHWDSSRPTKWF	22%	36%	-7 (FR1)/ +3 (FR3)
64L				ATYCHWDSSRPTKWF	22%	36%	-7 (FR1)/ +3 (FR3)
80L				ATYCHWDSSRPTKWF	19%	28%	-9 (FR1)/ +3 (FR3)
84L				DYTCMWDSSRLNWF	16%	28%	-4 (FR1)/ +3 (FR3)
86L				DYTCMWDSSRGSFWSF	21%	35%	-4 (FR1)/ +3 (FR3)
87L				DYTCMWDSSRGSFWSF	21%	33%	-4 (FR1)/ +3 (FR3)
109L				DYTCMWDSSRGSFWSF	17%	34%	-4 (FR1)/ +3 (FR3)
PGT121	IGHV3-21	IGHJ3*02	15	DYTCIWDSSRPTKWF	22%	28%	-7 (FR1)/ +3 (FR3)
PGT122				DYTCIWDSSRPTKWF	18%	26%	-7 (FR1)/ +3 (FR3)
PGT123				DYTCIWDSSRPTKWF	22%	34%	-7 (FR1)/ +3 (FR3)
PGT124				DYTCMWDSSRGSFWSF	19%	35%	-4 (FR1)/ +3 (FR3)
PGT133				DYTCMWDSSRPTKWF	26%	37%	-5 (FR1)/ +3 (FR3)
PGT134				DYTCMWDSSRPTKWF	27%	39%	-5 (FR1)/ +3 (FR3)

Figure 1. Mutation summary clones, including CDR3 sequences, mutation, and insertions/deletions (reproduced from Sok *et al.* [27]).

ture. Predicting neutralization breadth (the proportion of viral strains neutralized) directly from the antibody’s variable heavy-chain (V_H) sequence is challenging because the functional target depends on both heavy-chain contributions, antigen structure, and other factors. CATNAP aggregates measurements collected across studies and heterogeneous neutralization panels [28].

Protein language models (PLMs) utilize transformer architectures to learn contextual, high-dimensional sequence representations from unlabeled proteins. The multimodal PLM ESM3 learns through self-supervision to reason over protein sequence, 3D structure, and function [1, 12]. This raises two questions. First, can the general-protein representations that ESM3 learns support out-of-lineage bnAb modeling using only the variable heavy-chain amino-acid sequence? Here, “heavy-chain only” means the models re-

ceive no light-chain sequence or pairing, viral-envelope sequence, antigen structure, germline annotation, or assay-panel metadata. Second, does unsupervised tuning to antibody repertoires (using sequence-only data from the Observed Antibody Space [OAS] database [21]) improve this transfer?

We guard against data leakage because antibody datasets can contain many related variants; thus, a naive random split can place members of the same clonal lineage in both training and validation folds. This allows models to train on ancestors or descendants, rather than learning generalizable properties. In this project, we implement metadata-derived evaluation groups to isolate related antibodies across folds. Group labels control fold assignment and group-resampled uncertainty calculations; they are never supplied as model features. Because the study uses the same dataset to compare multiple model architectures (heads) and pooling methods, the cross-validation scores are exploratory.

We organize the study around three analyses. First, we define and assess a metadata grouping protocol and use resampling to quantify sensitivity to cohort composition for CATNAP classification and regression [28]. Second, we compare sequence features, frozen ESM3, and OAS-adapted ESM3 under matched folds and downstream heads. Third, we evaluate whole-sequence VAE reconstruction and a known-intermediate lineage interpolation test.

2. Related Work

2.1. HIV bnAb measurement and development

CATNAP aggregates antibody-by-virus neutralization measurements and associated metadata from published studies [28], including IC50 and IC80 measurements from virus neutralization assays across antibody-virus pairs and providing the consistent numerical labels for regression targets. Longitudinal studies of the CAP256/VRC26 lineage show that breadth can change substantially as variants accumulate and that somatically related members can span a wide range of neutralization breadth and potency [9, 10]. Additionally, we need to do lineage-held-out evaluation, lineage membership is a shortcut could be used instead of learning generalizable features.

2.2. Machine learning for HIV neutralization

Many HIV-bnAb works predict *virus* resistance to a fixed antibody panel rather than scoring antibody breadth from sequence; this review surveys this literature [5]. In this framing the model input is typically the amino-acid sequence or derived features of the HIV-1 envelope glycoprotein (Env) from the tested virus, and the output is a binary or continuous resistance label for a specific antibody. This virus-centered setup does not directly rank candidate antibodies by breadth. Recent virus-centered methods include

multi-task learning across antibodies, which mitigates HIV-1 subtype bias and captures co-resistance patterns shared across the antibody panel [14], and BRAVE (BnAb Resistance Analysis Via Evolutionary scale modeling 2), which uses ESM2 embeddings of full-length HIV-1 Env sequences with random-forest classifiers to predict resistance for 33 bnAbs [11]. By contrast, we evaluate an antibody-centered task using the V_H amino-acid sequence as the sole model input.

2.3. Protein and antibody language models

ESM2 showed that atomic-resolution structural information emerges in scaled protein-language-model representations, enabling sequence prediction [17]. ESM3 extended this approach to generative modeling structure, and function track as well [12]. In contrast, antibody models use sequence data tailored to the unusual statistics of the immunoglobulin variable regions. AntiBERTy was trained on hundreds of millions of natural antibody sequences, and separately used weak supervision to identify key binding residues [25]. AbLang trained separate models for heavy and light chains using OAS, reflecting the distinct mutational statistics of each chain, and demonstrated an advantage for sequence restoration [22]. More recent antibody models scale further, and target known biases: IgBert and IgT5 use over two billion unpaired and two million paired OAS sequences [15], while AbLang-2 investigates germline bias in antibody language models and methods intended to reduce that bias for antibody-design tasks [23]. Observed Antibody Space provides cleaned, annotated sequences used to train these models and used here for adaptation [15, 21, 22].

2.4. General versus domain-specific representations

Whether antibody-specific pretraining is needed is unclear. Across diverse immune tasks, general protein language models can match domain-specific models on many tasks, with results depending strongly on model size, available downstream data, and embedding-layer choice [7]. This result is non-obvious because antibody repertoires are shaped by V(D)J recombination and somatic hypermutation, creating distributions that differ from general protein corpora. Large general models may compensate through rich learned representations. We therefore compare ESM3 representations against explicit sequence-derived features on the functional task of neutralization breadth.

2.5. Adaptation and latent interpolation

Low-Rank Adaptation (LoRA) freezes base weights and learns low-rank updates in selected linear layers; our run targets attention and feed-forward projections [13]. We use a masked-language-modeling objective to adapt ESM3 on unlabeled OAS antibody sequences; no bnAb labels enter adaptation. Variational autoencoders (VAEs) optimize

a reconstruction objective together with a KL-divergence term that regularizes the posterior toward a prior distribution [16]. The regularization motivates testing whether linear paths in latent space correspond to smoother interpolations between data points. However, whether this holds depends on the representation, decoder fidelity, and the quantity and quality of training data. Our whole-sequence VAE experiment compares latent midpoint organization and decoded sequence recovery against the observed flanking sequences.

3. Data and Evaluation Protocol

3.1. CATNAP cohort and labels

We compile a dataset from the CATNAP database, which aggregates antibody sequences and quantitative viral neutralizations. The half-maximal inhibitory concentration (IC_{50}) measures an antibody’s potency against a specific viral isolate: it gives the concentration of antibody needed to neutralize 50% of the virus (lower values indicate higher antibody potency). An antibody–virus pair is counted as neutralized when $IC_{50} \leq 50 \mu\text{g/mL}$. Neutralization breadth b_i , for an antibody i , is the ratio of tested antibody–virus measurements neutralized below the threshold, also given here:

$$b_i = \frac{\#\{\text{measurements with } IC_{50} \leq 50 \mu\text{g/mL}\}}{\#\{\text{tested antibody–virus measurements}\}}, \quad (1)$$

We require at least 20 tested viruses to reduce sampling variability. For a binary classification task, we assign the “broad” label (1) where $b_i \geq 0.50$ and the narrow label (0) where $b_i \leq 0.20$. We exclude the middle band to establish a high-contrast classification task rather than creating a one-dimensional boundary with no area. This lets us classify breadth values as confidently broad or narrow. This is an analysis choice, not a biological or clinical cutoff: among 450 sequence-complete antibodies tested against at least 20 viruses, it excludes 150 middle-band antibodies and retains 300 labeled examples. We require a complete three complementarity-determining regions (CDRs) that form much of the antigen-binding surface. Figure 2 locates the region within the antibody structure.

The final labeled set contains the 300 unique V_H sequences (202 broad and 98 narrow), with sequence lengths ranging from 116 to 146 amino acids. For continuous-value breadth regression, we predict the ratio b_i directly for the 300 antibodies.

3.2. Lineage grouping and split

To prevent data leakage, we partition antibodies using metadata-derived groups. Groups are assigned sequentially from CATNAP clonal-lineage metadata (216 antibodies),

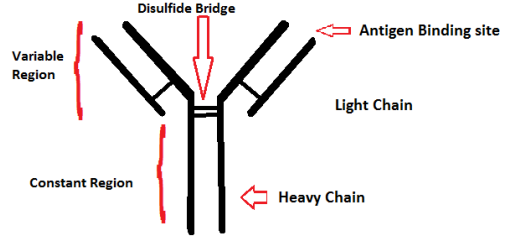


Figure 2. Schematic antibody structure highlighting heavy and light chains, variable and constant domains, the antigen-binding site. (Mamahdi14; cropped by Marbletan; CC BY-SA 4.0) [20].

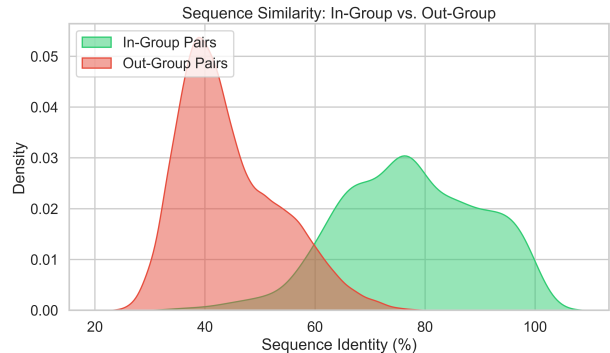


Figure 3. Pairwise Levenshtein sequence similarity for antibody pairs within the same group, and across groups. The overlap shows that our group method, as expected is not perfectly separable.

patient metadata indicating the same donor (70), single-linkage sequence clustering at $\geq 80\%$ Levenshtein identity (2), and individual singleton groups for the remainder (12). This hierarchical assignment first produces 113 conservative group IDs. We then merge connected group components linked by residual pairs at $\geq 80\%$ identity, producing 109 final groups; no cross-group near-duplicate pairs remain at this threshold. Across 862 in-group pairs, median pairwise sequence identity is 0.771 (Interquartile Range [IQR] 0.683–0.870); across 43,988 out-group pairs, the median is 0.424 (IQR 0.377–0.504).

Both classification and regression tasks utilize a 5-fold StratifiedGroupKFold split. The split keeps every metadata-derived group within one fold while approximately preserving binary-label balance across folds.

3.3. Metrics, selection, and uncertainty

We evaluate classification performance using the Area Under the Receiver Operating Characteristic curve (AUROC) and Area Under the Precision-Recall Curve (AUPRC, or average precision). AUROC measures the probability that the classifier ranks a random positive instance higher than a random negative one. AUPRC evaluates the precision–recall trade-off and is informative under class imbalances. For re-

gression on the continuous breadth metric, we report: the Pearson correlation coefficient (r), which measures linear correlation; the Spearman rank correlation (ρ), which evaluates monotonic but non-linear relationships; and the coefficient of determination (R^2), representing the proportion of target variance explained by the model compared to a mean baseline.

For hyperparameter tuning and configuration selection, we use the mean score across the 5 validation folds (mean AUROC for classification; mean Spearman ρ for regression). After selecting a configuration, we report the primary performance metrics on pooled out-of-fold (OOF) predictions (concatenating the held-out predictions from all folds). Pooled OOF metrics weight each held-out antibody once rather than weighting folds equally. We calculate 95% uncertainty intervals by resampling complete groups with replacement rather than individual sequences. To compute paired deltas that evaluate model differences, we resample the groups and calculate the metric difference on aligned OOF predictions. These intervals quantify sensitivity to the observed group composition; they do not account for configuration-selection bias or external-cohort variation.

3.4. OAS adaptation data

The executed domain-adaptation (masked-modeling) run uses 219,837 globally deduplicated human heavy-chain sequences from 11 OAS data units (108,675 clonotypes), with the largest unit contributing 20,000 sequences (9.1%). Whole data units that are systematically 5' truncated do not enter the corpus. A split produces 198,130 training and 21,707 validation sequences. OAS supplies the data used for adaptation [21]. We neither label those sequences as HIV bnAbs nor include them in downstream test folds.

For whole-sequence VAE pretraining, we protect every sequence that can later enter an evaluation. Applying exact-match and $\geq 80\%$ normalized Levenshtein-identity protection retains 169,876 of the 219,837 OAS candidates. A protected CATNAP adaptation set retains 143 of 568 candidate rows. OAS and the protected CATNAP background are training sources only; neither contributes evaluation rows.

3.5. Evaluation cohorts

Classification, regression, and whole-sequence VAE out-of-fold reconstruction use a common evaluation cohort comprising the same 300 antibody IDs, heavy-chain sequences, labels, and 109 metadata-derived groups. Classification and regression use matched folds. The VAE uses an independent five-fold grouped partition of this cohort while preserving the same group-level separation between training and evaluation data.

The known-intermediate lineage analysis constitutes a separate evaluation. It includes six VRC34 states drawn from the evaluation cohort and the lineage-trajectory data,

yielding four ordered triples. All six states are excluded from the shared OAS and CATNAP data used for VAE pre-training and adaptation.

4. Representations and Models

4.1. Baseline

The baseline represents each V_H residue tokens, and several meta-features. The meta-features are sequence length, and max protein token, that takes the value of the most common residue. The label is the continuous breadth value. These simple features test whether ESM3 adds signal beyond composition, proteins, sequence meta-data.

We evaluate four baselines: length-only logistic regression; logistic regression using all feature; random forest using all feature descriptors [2]; and XGBoost using all feature descriptors [3]. XGBoost is retained as the prespecified paired comparator for uncertainty analyses.

4.2. Frozen ESM3 and downstream heads

We represent each sequence with residue embeddings from the frozen, open-weight 1.4-billion-parameter `esm3-sm-open-v1`, trained on natural proteins with synthetic augmentation [1]. We pass each amino acid sequence through the model, extract the 1536-dimensional representation for each residue (excluding start-of-sequence BOS and end-of-sequence EOS tokens), resulting in a matrix of size $L \times 1536$ for a sequence of length L . To convert these variable-length representations into a fixed-size feature vector for downstream modeling, we compare two non-learned pooling schemes. Mean pooling averages each embedding coordinate across all residues, producing one 1536-dimensional vector that summarizes the sequence's typical representation. Mean+max+standard-deviation pooling computes, for every embedding coordinate, the average activation, strongest activation observed at any residue, and residue-to-residue variability, then concatenates the three summaries into a 4608-dimensional vector. Thus, the second mode preserves localized extremes that mean pooling discards. The downstream models apply normalization to each training fold. Classical linear, random-forest, MLP, scaling, and PCA components use `scikit-learn` [24]; XGBoost uses the `xgboost` package, and attention pooling uses `PyTorch`.

Classification candidates are L1/L2 logistic regression (using the SAGA solver [6] for L1), random forest [2], XGBoost, and a PCA-50 multilayer perceptron. A separately trained additive attention-pooling classifier is an additional capacity test. Regression candidates are Ridge, random forest [2], and XGBoost, with a length-only Ridge diagnostic baseline.

4.3. OAS LoRA adaptation

To test whether antibody-repertoire adaptation changes ESM3 representations, we perform domain adaptation using Low-Rank Adaptation (LoRA) [13]. LoRA freezes the base model parameters and injects trainable low-rank decomposition matrices into the transformer’s attention and feed-forward layers, reducing the trainable parameter count. We configure rank-16 adapters ($\alpha = 32$, dropout 0.05), training 18,874,368 parameters out of 1.42 billion total parameters (1.33%). Adaptation uses a self-supervised Masked Language Modeling (MLM) objective [8]: we randomly mask or corrupt 15% of amino acid tokens, and the model learns to predict the original residues from the surrounding context. We train for one epoch using the AdamW optimizer [19] with an effective batch size of 32. Performance evaluation uses cross-entropy loss and perplexity. Our run achieves a validation loss of 0.5025 and a perplexity of 1.653. We merge the learned adapter weights directly into the base model parameters before extracting embeddings.

4.4. Whole-sequence VAE and controls

To model each antibody as one coherent unit, we train a categorical VAE directly on complete heavy-chain sequences. We left-align each sequence in 170 slots and pad the remainder. Each slot is one-hot encoded over 20 amino acids plus a PAD token, producing a 3570-dimensional input. A multilayer encoder with widths 2048, 1024, and 256 maps the sequence to a 32-dimensional Gaussian latent variable; a mirrored decoder produces 170×21 categorical logits. The objective combines categorical reconstruction cross-entropy with Kullback–Leibler regularization and linear β warm-up. We bound posterior log-variance to $[-10, 6]$ and clip gradients to prevent non-finite training.

Training follows a protected staged protocol. The model first pretrains on the protected OAS background, adapts once on protected non-evaluation CATNAP sequences, and independently fine-tunes from the shared adapted checkpoint inside each canonical outer training fold. Inner group-held validation selects the epoch count; refitting then uses all eligible outer-training rows. Each canonical antibody receives exactly one OOF reconstruction and latent vector. We report residue reconstruction accuracy, categorical cross-entropy, and Levenshtein sequence edit distance.

For the known-intermediate lineage analysis, we evaluate four ordered VRC34 triples (A, B, C) in which B is a known inferred intermediate. We encode each sequence to its posterior mean and form the midpoint

$$z_{\text{mid}} = \frac{1}{2}(\mu_A + \mu_C). \quad (2)$$

The latent test asks whether z_{mid} is closer to μ_B than either endpoint. The sequence test compares the decoded midpoint with B , the decoded endpoints, and the raw endpoint

Representation	Fold AUROC	Pooled AUROC	95% interval
XGB	0.660 ± 0.087	0.651	[0.559, 0.758]
Frozen ESM3	0.779 ± 0.128	0.773	[0.692, 0.853]
OAS LoRA ESM3	0.746 ± 0.129	0.729	[0.639, 0.818]

Table 1. Classification performance for the selected baseline, frozen, and LoRA ESM3 configurations. A group bootstrap produces the intervals.

sequences. We run this test after OAS pretraining, CATNAP adaptation, and the fold that holds out the complete VRC34 group.

Cross-representation midpoint controls. *New addition (2026-06-19).* To compare the VAE latent midpoint against the foundation model on the same four triples, we add frozen-ESM3 and LoRA-ESM3 embedding-interpolation controls: for each triple we globally align the per-residue embeddings of A and C , form the embedding midpoint, and measure its distance to B . Because raw distances are on incomparable scales across the VAE latent, frozen ESM3, and LoRA ESM3, we report all arms after per-dimension standardization (dividing each dimension by its own standard deviation, yielding unit variance) — the same standardization rationale applied to the classifier embeddings. This makes the midpoint errors directly comparable across representations (Sec. 5.5).

Sequence-VAE latent as features. *New addition (2026-06-19).* We also evaluate the whole-sequence VAE’s 32-dimensional latent as a discriminative feature vector. To stay leakage-safe, each canonical antibody is encoded by the fold checkpoint that held its lineage out (its out-of-fold posterior mean); we then apply the same classification and regression heads and grouped cross-validation as the ESM3 representations (Sec. 5.7). The PCA-50 multilayer perceptron head is omitted for this arm because its component count exceeds the 32 latent dimensions.

5. Results

5.1. Classification

Table 1 separates the configuration-selection score (mean over folds) from the pooled OOF estimate and group-bootstrap interval. Table 2 reports all four candidates. Random forest has the largest mean-fold AUROC (0.671), while the prespecified XGBoost comparator has the largest pooled OOF AUROC (0.651) and AUPRC (0.771). Against that XGBoost comparator, Frozen ESM3 raises the pooled AUROC point estimate from 0.651 to 0.773 and pooled AUPRC from 0.771 to 0.869. On aligned predictions, the paired frozen-minus-base difference is $+0.122$ AUROC ($[-0.008, 0.240]$) and $+0.099$ AUPRC ($[-0.003, 0.189]$).

Model	Features	Fold AUROC	Pooled AUROC	Pooled AUPRC
Logistic regression	Length only	0.614 $\pm$ 0.104	0.581	0.720
Logistic regression	All seven	0.574 $\pm$ 0.139	0.567	0.683
Random forest	All seven	0.671 $\pm$ 0.089	0.639	0.762
XGBoost	All seven	0.660 $\pm$ 0.087	0.651	0.771

Table 2. Classification baseline results. Fold AUROC is the mean and standard deviation across five grouped folds; pooled metrics concatenate all OOF predictions. “All seven” denotes the complete handcrafted heavy-chain feature set. Bold marks the largest value in each metric column; XGBoost is the prespecified comparator used for the paired uncertainty analysis.

Both intervals include zero. The intervals describe sensitivity to held-out lineage composition; they do not remove model-selection optimism.

LoRA’s selected classifier reaches pooled AUROC 0.729, below frozen by -0.044 with paired interval $[-0.108, 0.019]$. The corresponding AUPRC difference is -0.051 $[-0.090, -0.010]$. Thus, this single adaptation run does not improve classification and reduces selected-model AUPRC on the observed cohort. The receiver operating characteristic and precision–recall curves for these classifiers appear in Figure 4.

The learned attention pool reaches mean-fold AUROC 0.769 ± 0.117 , below the best fixed-pooling frozen model. Adding learned attention pooling did not improve mean-fold AUROC on this cohort. Table 3 reports all fixed-pooling candidates.

5.2. Continuous breadth regression

Regression uses the same 300 threshold-selected antibodies. As summarized in Table 4, frozen ESM3 reaches pooled Spearman $\rho = 0.368$ $[(0.157, 0.574)]$, Pearson $r = 0.363$ $[(0.163, 0.564)]$, and $R^2 = 0.107$ $[-0.122, 0.307]$. The length-only pooled Spearman diagnostic baseline is 0.007. Thus the selected representation shows a positive pooled ranking association, while its absolute breadth predictions remain weak and sensitive to group composition. The scatter plots of predicted versus true breadth values appear in Figure 5, illustrating the prediction variance.

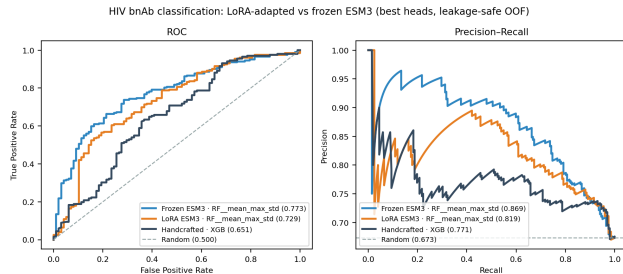


Figure 4. Pooled OOF ROC and precision–recall curves for the selected classifiers.

Configuration	Frozen		LoRA	
	AUROC	AUPRC	AUROC	AUPRC
RF, mean+max+std	0.779	0.878	0.746	0.854
RF, mean	0.761	0.863	0.732	0.842
XGB, mean+max+std	0.755	0.861	0.671	0.781
L1 LR, mean	0.753	0.857	0.734	0.835
L1 LR, mean+max+std	0.735	0.856	0.712	0.817
XGB, mean	0.733	0.839	0.706	0.814
L2 LR, mean+max+std	0.720	0.846	0.662	0.791
MLP, mean+max+std	0.699	0.821	0.637	0.742
L2 LR, mean	0.697	0.830	0.728	0.829
MLP, mean	0.696	0.831	0.703	0.787

Table 3. Mean grouped-fold classification results for all frozen and LoRA fixed-pooling candidates. Bold marks the frozen and LoRA configurations selected by mean-fold AUROC.

Representation	Fold ρ	Pooled ρ	95% interval
Frozen ESM3	0.380 $\pm$ 0.319	0.368	$[0.157, 0.574]$
OAS LoRA ESM3	0.311 $\pm$ 0.292	0.305	$[0.052, 0.549]$
Length only	—	0.007	—

Table 4. Continuous-breadth regression performance. Pooled OOF metrics and group-bootstrap intervals evaluate the selected configurations.

Selecting LoRA configurations by mean-fold Spearman chooses random forest with mean pooling. Its pooled Spearman is 0.305 $[(0.052, 0.549)]$. The paired LoRA-minus-frozen Spearman difference is -0.062 $[-0.201, 0.082]$.

5.3. Whole-sequence VAE reconstruction

Across the same canonical 300-antibody cohort used by classification and regression, the whole-sequence VAE attains mean OOF residue reconstruction accuracy 0.642, categorical cross-entropy 1.348, and sequence edit distance 43.35. Reconstruction varies across the five grouped folds: mean residue accuracy ranges from 0.563 to 0.688. Broad/positive antibodies are harder to reconstruct than narrow/weak antibodies, with mean residue accuracy 0.597 versus 0.735. Thus, the model learns substantial sequence regularity but does not provide high-fidelity OOF recon-



Figure 5. Pooled OOF predicted versus observed breadth for frozen and LoRA configurations.

Stage	Latent recoveries	Latent midpoint	Decoded midpoint edit	Raw endpoint edit
OAS pretrained	2/4	0.170	24.75	5.00
CATNAP adapted	2/4	0.211	29.25	5.00
VRC34-held-out fold	2/4	0.219	28.00	5.00

Table 5. Whole-sequence VAE known-intermediate results over four VRC34 lineage triples. Latent midpoint is the mean distance to the known intermediate; lower is better. Edit columns report amino-acid Levenshtein distance. Bold marks the lowest decoded-midpoint mean, not a successful comparison against raw endpoints.

struction.

5.4. Known-intermediate lineage recovery

Table 5 reports the exact four-triple VRC34 midpoint test. At every stage, the latent midpoint is closer to the known intermediate than both latent endpoints in two of four triples. This is partial latent-geometry evidence, not consistent recovery across the trajectory.

Decoded sequence recovery is unsuccessful under the stronger control. No decoded midpoint is closer to the known intermediate than the nearer raw flanking sequence. The OAS-pretrained model has the lowest decoded-midpoint mean edit distance (24.75); CATNAP adaptation and fold fine-tuning do not improve it. Appendix A.8 lists the six VRC34 triple sequences and our generated decoded midpoints, with their differences from the known intermediate highlighted. validation loss is still improving, so this adaptation stage is cap-limited rather than demonstrably converged.

(Note, added 2026-06-19: the latent-midpoint distances in Table 5 are VAE-internal and live on a different scale than the ESM3 embedding distances, so they cannot be compared across representations directly. The standardized cross-representation comparison in Table 6 below addresses this.)

5.5. Standardized cross-representation midpoint comparison

New analysis (added 2026-06-19); supplements the VAE-only table above and is not part of the original audited tables. To place the antibody-trained VAE next to the foundation model on identical triples, we add frozen-ESM3 and LoRA-ESM3 embedding-interpolation controls and report every arm after per-dimension standardization, so the three representations share a common unit-variance scale (raw midpoint distances differ by orders of magnitude — VAE latent ≈ 0.02 , frozen ESM3 ≈ 11 , LoRA ESM3 ≈ 14 — and are not comparable). Table 6 reports the CATNAP-adapted stage over the four VRC34 triples. The sequence VAE and frozen ESM3 each place the midpoint nearer the known intermediate than both endpoints in two of four triples; the LoRA-ESM3 midpoint beats its own endpoints in all four,

Representation	Std. midpoint MAE	Recovers B
Sequence VAE (latent)	0.103	2/4
Frozen ESM3	0.085	2/4
LoRA ESM3	0.108	4/4

Table 6. Added 2026-06-19. Standardized (per-dimension z-scored) midpoint distance to the known VRC34 intermediate B and the number of triples in which the midpoint beats both endpoints, for the CATNAP-adapted stage. Standardizing each representation to unit per-dimension variance makes the three arms comparable; raw distances (not shown) span orders of magnitude. Bold marks the smallest mean distance and the most consistent recovery, not a statistically tested winner.

although its standardized midpoint distance is not the smallest. With four triples from a single lineage and no confidence intervals, this is descriptive only.

5.6. Decoded-midpoint sequence recovery

New analysis (added 2026-06-19). The standardized comparison above is in representation space. To compare in *sequence* space, we decode each midpoint to amino acids and measure edit distance to the true intermediate B . Edit distance (amino-acid Levenshtein to the same target B) is a common, model-agnostic sequence-space scale, so — unlike the representation-space comparison in Table 6 — no per-dimension standardization is required. The frozen-ESM3 and LoRA-ESM3 embedding midpoints are decoded with the frozen ESM3 sequence head (`output_heads.sequence_head`); this head is frozen for both models — LoRA adapts only attention/feed-forward layers and the masked-LM objective used this same head as readout — so it is the matched decoder for each. The VAE midpoint is decoded by the VAE’s own head (CATNAP-adapted stage, matching Table 6). As Table 7

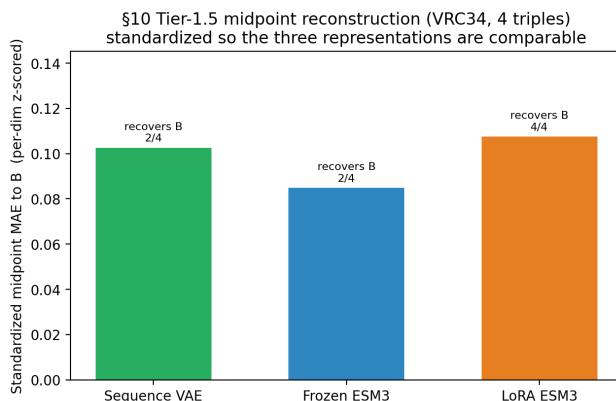


Figure 6. Added 2026-06-19. Standardized midpoint distance to the known VRC34 intermediate across the three representations (recovers- B count annotated).

Triple ($\rightarrow B$)	Frozen ESM3	LoRA ESM3	Seq. VAE	Raw endpoint
$\rightarrow p11$	8	2	19	2
$\rightarrow p12$	17	11	26	9
$\rightarrow p13$	23	17	34	5
$\rightarrow p14$	27	19	38	4
Mean	18.75	12.25	29.25	5.00

Table 7. *Added 2026-06-19.* Decoded-midpoint edit distance to the true VRC34 intermediate B . Frozen- and LoRA-ESM3 embedding midpoints are decoded with the frozen ESM3 sequence head; the VAE latent midpoint by its own decoder (CATNAP stage). Raw endpoint is the nearer flanking sequence (lower is better). The ESM3 decoders recover B far better than the VAE, but none beats the raw-endpoint control on average.

shows, decoding the ESM3 embedding midpoints recovers the intermediate markedly better than the VAE (mean edit distance: LoRA 12.25, frozen 18.75, versus VAE 29.25), and LoRA matches the raw-endpoint baseline on the first triple. None beats the raw-endpoint baseline on average (5.00), so midpoint interpolation still does not surpass simply taking the nearer flanking sequence; but the ESM3 decoders are far closer to the intermediate than the VAE.

5.7. Sequence-VAE latent as a discriminative representation

New analysis (added 2026-06-19); supplements the classification (Table 1) and regression (Table 4) results above. We also test the whole-sequence VAE’s 32-dimensional latent as a discriminative feature. To remain leakage-safe, each canonical antibody is encoded by the fold checkpoint that held its lineage out (its out-of-fold posterior mean), and the same heads and grouped cross-validation as the ESM3 representations are applied. As Table 8 and Figure 7 show, the latent’s best classifier reaches mean-fold AUROC 0.682 and its best regressor pooled Spearman 0.114, both well below frozen ESM3 (0.779 / 0.368) and LoRA ESM3 (0.746 / 0.305). On this low-dimensional latent, linear heads outperform tree ensembles (random forest 0.608, XGBoost 0.606 AUROC), reversing the pattern on the high-dimensional ESM3 embeddings. The 32-dimensional reconstruction bottleneck therefore retains the least breadth-discriminative signal, though it remains above chance.

6. Limitations

6.1. Exploratory model selection

We compare heads and pooling schemes on the same five grouped folds. Selecting the best mean-fold configuration makes its reported Cross-validation performance may be optimistic due to hyperparameter and model-selection bias. A future confirmatory evaluation requires nested grouped cross-validation.

Representation	AUROC (mean-fold)	Spearman (pooled)
Frozen ESM3	0.779	0.368
LoRA ESM3	0.746	0.305
Sequence-VAE latent	0.682	0.114

Table 8. *Added 2026-06-19.* Best-head discriminative performance of the three representations on the canonical 300/109 leakage-safe split: classification mean-fold AUROC and breadth-regression pooled Spearman. Frozen and LoRA values match Tables 3 and 4; the sequence-VAE-latent row is new.

6.2. Cohort and target construction

The cohort contains only 300 sequences across 109 metadata-derived groups, limiting the statistical power of our architectural comparisons. Models use V_H sequences without V_L sequences or pairing, omitting light-chain and the heavy-light-interface determinants of antibody structure and function. CATNAP narrow/weak antibodies are not equivalent to arbitrary antibodies our metric of neutralization breadth is a simple, unweighted fraction over viral panels that vary in size and strain composition, without accounting for laboratory-specific or panel-level effects.

6.3. Grouping assumptions

Metadata-derived clonal lineage is the strongest available grouping key. Patient-level groups may merge unrelated antibodies from the same donor. The sequence-identity clustering heuristic ($\geq 80\%$ identity) supports separation, but cannot biologically rule out developmental independence.

6.4. Adaptation and ESM3 constraints

We evaluate the LoRA domain adaptation experiment on a single corpus with one adapter configuration and one training run. This experiment cannot determine whether other adaptation corpora, objectives, configurations, or repeated runs would improve downstream performance. Additionally, using a 1.4B parameter model like ESM3 introduces substantial computational overhead for inference and adaptation compared with simpler sequence-based models.

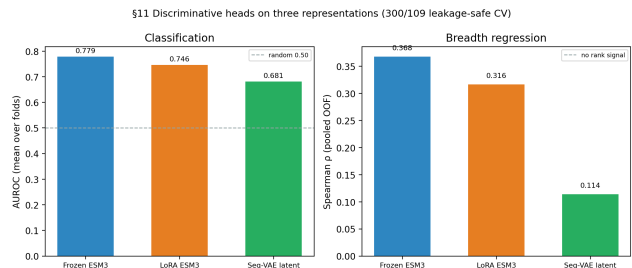


Figure 7. *Added 2026-06-19.* Classification AUROC (mean over folds) and breadth-regression pooled Spearman for the three representations.

6.5. Interpolation and VAE constraints

The whole-sequence VAE preserves antibody-level sequence context, but its left-aligned slots do not use biological IMGT/ANARCI numbering; insertions and deletions can therefore shift downstream positions. OOF reconstruction remains moderate, especially for broad antibodies. The known-intermediate analysis contains only four ordered triples from one inferred VRC34 trajectory, so it is a case study rather than a population estimate. Raw flanking sequences are naturally close to their known intermediate, which creates a strong and appropriate control. Finally, the model uses one random seed, and CATNAP adaptation reaches its epoch cap while validation loss is still improving.

7. Conclusion

Under metadata-derived grouped evaluation, frozen ESM3 yields higher classification point estimates than the baseline XGBoost comparator and a positive pooled association with continuous breadth. However, paired classification intervals include zero and pooled R^2 remains low. Thus, the evaluated V_H -only models do not provide accurate absolute breadth prediction on this cohort. The single OAS LoRA adaptation run does not improve downstream classification or regression performance.

The evaluated task differs from virus-centered resistance studies that score Env sequences against fixed antibodies [5, 11, 14] and from an antibody-centered structure-based breadth estimator for CD4-binding-site antibodies [4]. Here, the model receives the variable heavy-chain amino-acid sequence as its sole input, without light-chain sequence, viral-envelope sequence, epitope labels, assay-panel metadata, or modeled structure. Because these studies use different inputs, targets, and split protocols, their AUROCs are not directly comparable. Against the prespecified baseline XGBoost comparator, frozen ESM3 increases pooled AUROC from 0.651 to 0.773, but the paired +0.122 difference has group-bootstrap interval $[-0.008, 0.240]$.

The whole-sequence VAE reconstructs 64.2% of held-out residues on average. Its latent midpoint recovers the known VRC34 intermediate in two of four lineage triples, but decoded midpoints do not outperform the nearer raw endpoint. This supports partial latent organization, not successful sequence-level intermediate recovery.

Overall, metadata-derived grouping, matched evaluation, pooled OOF reporting, and group-resampled intervals materially change the interpretation of these experiments. Future work should evaluate paired heavy/light-chain inputs and structural context, use larger external cohorts, repeat training across seeds, and replace left alignment with biologically numbered sequence positions.

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A. Verification and Reproducibility Details

A.1. Candidate configurations

A.2. Classification detail

Table 9 reconciles the configuration selection score with the pooled estimates used in the main text.

Representation	AUROC	AUPRC	Fold AUROC
XGB	0.651	0.771	0.660 ± 0.087
Frozen ESM3 RF	0.773	0.869	0.779 ± 0.128
OAS LoRA ESM3 RF	0.729	0.819	0.746 ± 0.129

Table 9. Classification metrics reconciled from aligned OOF predictions. Frozen and LoRA RF use mean+max+std pooling. Configurations maximize mean-fold AUROC.

A.3. Regression detail

The fixed, mean-fold-selected regression configurations give the pooled estimates in Table 10.

Representation	Spearman	Pearson	R^2
Frozen ESM3	0.368	0.363	0.107
OAS LoRA ESM3	0.305	0.310	0.046
Length-only Ridge	0.007	-0.031	-0.037

Table 10. Pooled OOF breadth-regression metrics. Frozen and LoRA configurations maximize mean-fold Spearman.

A.4. Evaluation cohort provenance

Table 11 summarizes the cohorts used for each reported evaluation. Primary out-of-fold analyses share one harmonized CATNAP cohort, whereas the known-intermediate analysis augments that cohort only with the specified VRC34 lineage states.

Analysis	Evaluation cohort	Scope
Classification	Harmonized CATNAP	300 antibodies / 109 groups
Regression	Harmonized CATNAP	same 300 / 109
Sequence-VAE OOF	Harmonized CATNAP	same 300 / 109; independent folds
Sequence-VAE known intermediate	CATNAP + VRC34 trajectories	4 triples / 6 states

Table 11. Evaluation cohorts used in the reported analyses. OAS and protected non-evaluation CATNAP sequences are used only for model training.

A.5. Per-configuration discriminative results

Table 12 reports every fixed-pooling regression configuration compared for the frozen and LoRA representations. The entries are not independent experiments: all candidates reuse the same five reporting folds. Selection uses the mean-fold column, even when a different configuration has a larger pooled estimate. The complete classification performance across all heads and pooling schemes appears

in Figure 8, while Figure 9 details the regression Spearman correlations.

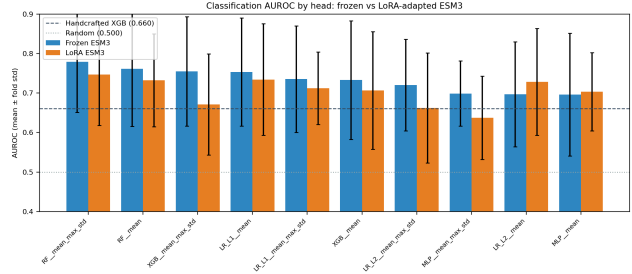


Figure 8. Mean grouped-fold AUROC (with standard deviation) for all evaluated classifier heads and pooling configurations under frozen and LoRA representations.

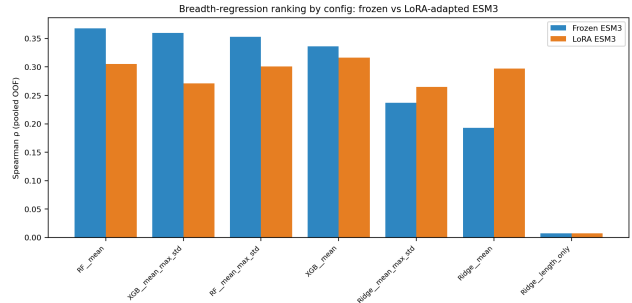


Figure 9. Regression Spearman correlation across different model configurations, comparing frozen and LoRA representations.

Configuration	Frozen Spearman		LoRA Spearman	
	Fold	Pooled	Fold	Pooled
RF, mean	0.380	0.368	0.311	0.305
RF, mean+max+std	0.364	0.353	0.307	0.301
XGB, mean+max+std	0.355	0.360	0.279	0.271
XGB, mean	0.326	0.336	0.310	0.316
Ridge, mean+max+std	0.241	0.237	0.255	0.265
Ridge, mean	0.157	0.193	0.278	0.297
Length-only Ridge	0.133	0.007	0.133	0.007

Table 12. Mean-fold and pooled-OOF regression Spearman correlations for all candidates. Bold marks the configurations selected by mean-fold Spearman.

A.6. Paired resampling results

Predictions are aligned by antibody ID, and complete metadata-derived groups are resampled as units. Table 13 gives the resulting paired differences and their group-bootstrap intervals.

A.7. Whole-sequence VAE fold summary

The sequence VAE evaluates each canonical antibody exactly once OOF. The independent grouped folds preserve the canonical group leakage boundary.

Comparison	Metric	Δ	Low	High
Frozen-baseline	AUROC	0.122	-0.008	0.240
Frozen-baseline	AUPRC	0.099	-0.003	0.189
LoRA-frozen	AUROC	-0.044	-0.108	0.019
LoRA-frozen	AUPRC	-0.051	-0.090	-0.010
LoRA-frozen	Spearman	-0.062	-0.201	0.082
LoRA-frozen	Pearson	-0.053	-0.215	0.102
LoRA-frozen	R^2	-0.062	-0.242	0.099

Table 13. Paired group-bootstrap differences for the selected configurations. Intervals quantify sensitivity to the observed metadata-derived groups and do not correct for configuration selection.

Fold	OOF rows	Residue accuracy	Mean edit distance
0	72	0.602	48.46
1	92	0.673	39.98
2	61	0.688	38.54
3	36	0.650	41.72
4	39	0.563	50.90

Table 14. Whole-sequence VAE OOF reconstruction by grouped fold.

A.8. Tier-1.5 VRC34 sequences and all generated midpoints

The four known-intermediate triples use the six VRC34 lineage states below (heavy-chain V_H , gap-stripped).

VRC34-UCA (122 aa)
 QVQLVQSGAEVKKPGASVKVSKASGYTFTGYMHWRQAPG
 QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSISTAYMELS
 RLRSDDTAIYYCARDKYYGNEAVGMDVWGQGTITVTVSS

VRC34-pI1 (122 aa)
 QVQLVQSGAEVKKPGASVKVSKASGYTFTGYMHWRQAPG
 QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSISTAYMEVS
 RLTSDDTAIYYCARDKYYGNEAVGMDVWGQGTITVTVSS

VRC34-pI2 (122 aa)
 QEVLVQSGAEVKKPGASVKVSKASGYTFTGHHMHWRQAPG
 QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSINTAYLDVT
 RLTSDDTAIYYCARDKYYGNEAVGMDVWGQGTITVTVSS

VRC34-pI3 (121 aa)
 QEVLVQSGAEVKKPGASVKVSKAFGYTFTGNPMHWRQAPG
 QGLEWMGWINPHSGDTTTSQKFQGRVYMTDRKSIINTAYLDVT
 RLTSDDTAIYYCARDKYYGNEAVGMDVWGQGTITVTVS

VRC34-pI4 (121 aa)
 QEVLVQSGAEVKKPGASVKVSCRAFGYFTGNALHWRQAPG
 QGLEWLGINPHSGDTTTSQKFQGRVYMTDRKSIINTAYLDVT
 RLTSDDTAIYYCARDKYYGNEAVGMDVWGQGTITVTVS

VRC34.01 (122 aa)
 QEVLVQSGAEVKKPGASVKVSCRAFGYFTGNALHWRQAPG
 QGLEWLGINPHSGDTTTSQKFQGRVYMTDRKSIINTAFLDVT
 RLTSDDTGIYYCARDKYYGNEAVGMDVWGQGTISVTVSS

For each triple (A, B, C) we show the true intermediate B aligned against every decoded midpoint we generated: the frozen-ESM3 and LoRA-ESM3 embedding midpoints decoded by the frozen ESM3 sequence head (frzn, lora), and the whole-sequence VAE latent midpoint decoded by its own head at the OAS, CATNAP, and fold stages (vOAS, vCAT, vfld). Residues differing from B are red; the per-row edit distances (amino-acid Levenshtein to B , a common scale) are in the header. Across all models and stages the differences concentrate in CDR3, the ESM3 decoders

land closer to B than the VAE, and none beats the nearer raw-endpoint baseline.

Triple 1: VRC34-UCA $\rightarrow$ VRC34-pI1 $\rightarrow$ VRC34-pI2. Edit distance to B : frozen 8, LoRA 2, VAE(OAS/CAT/fold) 16/19/18; raw endpoint 2.

```

B      QVQLVQSGAEVKKPGASVKVSKASGYTFTGYMHWRQAPG
frzn   QVQLVQSGAEVKKPGASVKVSKASGYTFTSYMHWRQAPG
lora    QVQLVQSGAEVKKPGASVKVSKASGYTFTGYMHWRQAPG
vOAS    QVQLVQSGAEVKKPGASVKVSKASGYTFTDYYIHWVRQAPG
vCAT    QVQLVQSGAEVKKPGASVRSVSKASGYTFTDYYIHWVRQAPG
vfld    QVQLVQSGAEVKKPGASVRSVSKASGYTFTDYYIHWVRQAPG

B      QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSISTAYMEVS
frzn   QGLEWMGWINPNRGGTNYAQKFQGRVTMTRDTSISTAYMELS
lora    QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSISTAYMELS
vOAS    QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSISTAYMELS
vCAT    QGLEWMGWINPNRGGTNYAQKFQGRVTMTRDTSISTAYMELT
vfld    QGLEWMGWINPNRGATNYAQKFQGRVTMTRDTSISTAYMELR

B      RLTSDDTAIYYCAR-----D-KYYGNEAVGMDVWGQGTITVTV
frzn   RLRSDDTAIYYCAR-----D-RYYGSYAYGMDVWGQGTITVTV
lora    RLRSDDTAIYYCAR-----D-KYYGNEAVGMDVWGQGTITVTV
vOAS    RLRSDDTAIYYCAR-----G-RRVGSYGYFDYWGQGTITVTV
vCAT    RLRSDDTAIYYCARGRLTSDYLYY-----MDYWGQGTITVTV
vfld    RLRSDDTAIYYCAR-----D-RSYGDYDWFYDYGQGTITVTV

B      SS
frzn   SS
lora    SS
vOAS    SS
vCAT    SS
vfld    SS

```

Triple 2: VRC34-pI1 $\rightarrow$ VRC34-pI2 $\rightarrow$ VRC34-pI3. Edit distance to B : frozen 17, LoRA 11, VAE(OAS/CAT/fold) 24/26/26; raw endpoint 9.

```

B      QEVLVQSGAEVKKPGASVKVSKASGYTFTGHHMHWRQAPG
frzn   QVQLVQSGAEVKKPGASVKVSKASGYTFTSYMHWRQAPG
lora    QVQLVQSGAEVKKPGASVKVSKASGYTFTGYMHWRQAPG
vOAS    QVQLVQSGAEVKKPGASVKVSKASGYTFTDYYIHWVRQAPG
vCAT    QVQLVQSGAEVKKPGASVRSVSKASGYTFTDYYIHWVRQAPG
vfld    QVQLVQSGAEVKKPGASVRSVSKASGYTFTDYYIHWVRQAPG

B      QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSINTAYLDVT
frzn   QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSISTAYMELS
lora    QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSISTAYMELS
vOAS    QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSISTAYMELS
vCAT    QGLEWMGWINPNRGGTNYAQKFQGRVTMTRDTSISTAYMELT
vfld    QGLEWMGWINPNRGATNYAQKFQGRVTMTRDTSISTAYMELR

B      RLTSDDTAIYYCAR-----D-KYYGNEAVGMDVWGQGTITVTV
frzn   RLTSDDTAIYYCAR-----D-RYYGSYAYGMDVWGQGTITVTV
lora    RLRSDDTAIYYCAR-----D-KYYGNEAVGMDVWGQGTITVTV
vOAS    RLRSDDTAIYYCAR-----G-RRVGSVLYYMDYWGQGTITVTV
vCAT    RLRSDDTAIYYCARGRRTSDYLYY-----MDYWGQGTITVTV
vfld    RLRSDDTAIYYCAR-----D-RSVGDYDYYMDYWGQGTITVTV

B      SS
frzn   SS
lora    SS
vOAS    SS
vCAT    SS
vfld    SS

```

Triple 3: VRC34-pI2 $\rightarrow$ VRC34-pI3 $\rightarrow$ VRC34-pI4. Edit distance to B : frozen 23, LoRA 17, VAE(OAS/CAT/fold) 28/34/32; raw endpoint 5.

```

B      QEVLVQSGAEVKKPGASVKVSKAFGYTFTGNPMHWRQAPG
frzn   QVQLVQSGAEVKKPGASVKVSKASGYTFTSYMHWRQAPG
lora    QVQLVQSGAEVKKPGASVKVSKASGYTFTGYMHWRQAPG
vOAS    QVQLVQSGAEVKKPGASVKVSKASGYTFTDYYIHWVRQAPG
vCAT    QVQLVQSGAEVKKPGASVRSVSKASGYTFTDYYIHWVRQAPG

```

```

vfld QVQLVQSGAEVKKPGASVRSCKASGYTFTDYYIHWRQAPG
B QGLEWMGWINPHSGDTTTAQKFQGRVYMTDRKSIINTAYLDVT
frzn QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSISTAYMELS
lora QGLEWMGWINPNSGGTNYAQKFQGRVTMTRDTSISTAYMELT
vOAS QGLEWMGWINPKSGGTNYAQKFQGRVTMTRDTSISTAYMELT
vCAT QGLEWMGWINPNTGGTNYAQKFQGRVTMTRDTSISTAYMELT
vfld QGLEWMGWINPNSGATNYAQKFQGRVTMTRDTSISTAYMELR
B RLTSDDTAIYYCAR----DKY-YGNEAVGMDVWGQGTITVTV-
frzn RLTSDDTAIYYCAR----DKY-YGNEAVGMDVWGQGTITVTVS
lora RLTSDDTAIYYCAR----DKY-YGNEAVGMDVWGQGTITVTVS
vOAS RLRSDDTAIYYCAR----GRR-VGSVLYYMDVWGQGTITVTVS
vCAT RLRSDDTAIYYCARGRRTDDYLY-----YMDYWGQGLVTVS
vfld RLRSDDTAIYYCAR----DRS-TGDYLYMDYWGQGLVTVS
B S
frzn S
lora S
vOAS S
vCAT S
vfld S

```

Triple 4: VRC34-pI3 → VRC34-pI4 → VRC34.01.

Edit distance to *B*: frozen 27, LoRA 19, VAE(OAS/CAT/fold) 31/38/36; raw endpoint 4.

```

B QEVLVQSGAEVKKPGASVKVSCRAFGYTFTGNALHWVRQAPG
frzn QVQLVQSGAEVKKPGASVKVSCASGYTFTSYMHWRQAPG
lora QVQLVQSGAEVKKPGASVKVSCASGYTFTGYMHWRQAPG
vOAS QVQLVQSGAEVKKPGASVKVSCASGYTFTDYYIHWRQAPG
vCAT QVQLVQSGAEVKKPGASVRSCKASGYTFTDYYIHWRQAPG
vfld QVQLVQSGAEVKKPGASVRSCKASGYTFTDYYIHWRQAPG
B QGLEWLGWINPHSGDTTTSQKFQGRVYMTDRKSIINTAYLDVT
frzn QGLEWMGWINPYSGGTKYAQKFQGRVTMTRDTSISTAYMELS
lora QGLEWMGWINPNSGDTKYAQKFQGRVTMTRDTSISTAYMELT
vOAS QGLEWMGWINPKSGGTNYAQKFQGRVTMTRDTSISTAYMELT
vCAT QGLEWMGWINPNTGGTNYAQKFQGRVTMTRDTSISTAYMELT
vfld QGLEWMGWINPNSGATNYAQKFQGRVTMTRDTSISTAYMELR
B RLTSDDTAIYYCAR-----D-KYYGNEAVGMDVWGQGTITVTV
frzn RLTSDDTAIYYCAR-----D-RYYGSAVAVGMDVWGQGTITVTV
lora RLTSDDTAIYYCAR-----D-KYYGNEAVGMDVWGQGTITVTV
vOAS RLRSDDTAIYYCAR-----G-RRVGSVLYYMDVWGQGTITVTV
vCAT GLRSDDTAIYYCARGRRTSDYLYY-----MDYWGQGLVTV
vfld GLRSDDTAIYYCAR-----D-RSTGDYDWMYMDYWGQGLVTV
B -S
frzn SS
lora SS
vOAS SS
vCAT SS
vfld SS

```

A.9. Reproducibility limits

The reported calculations can be reconstructed from saved OOF predictions and evaluation artifacts without retraining ESM3, the LoRA adapter, or the whole-sequence VAE. These artifacts support verification of internal consistency and reported calculations, but not bitwise reproduction of the GPU training runs. Similarly, the group bootstrap re-samples the observed 109 metadata-derived groups. It does not represent a new external cohort, and its intervals remain conditional on the selected model configurations.

Confirmatory evidence would require a locked external cohort or nested grouped model selection, repeated adapter and VAE seeds, biologically numbered sequence alignment, and additional known-intermediate lineages. These experiments remain outside the scope of the present study and define important directions for future validation.